```
# Task 3

# This program adds two numbers

num1 = 1.5
num2 = 6.3

# Add two numbers
sum = num1 + num2

# Display the sum
print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))
```

```
The sum of 1.5 and 6.3 is 7.8
```

```
# Task 4

# Store input numbers
num1 = input('Enter first number: ')
num2 = input('Enter second number: ')

# Add two numbers
sum = float(num1) + float(num2)

# Display the sum
print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))
```

```
Enter first number: 4
Enter second number: 5
The sum of 4 and 5 is 9.0
```

```
# Task 5

# Python Program to calculate the square root

# Note: change this value for a different result
num = 8

# To take the input from the user
#num = float(input('Enter a number: '))

num_sqrt = num ** 0.5
print('The square root of %0.3f is %0.3f'%(num
,num_sqrt))
```

```
The square root of 8.000 is 2.828
```

```
# Task 6

# Find square root of real or complex numbers
# Importing the complex math module
import cmath
```

```
num = 1+2j

# To take input from the user
#num = eval(input('Enter a number: '))

num_sqrt = cmath.sqrt(num)
print('The square root of {0} is {1:0.3f}+{2:0.3f}j'.format(num
,num_sqrt.real,num_sqrt.imag))
```

```
The square root of (1+2j) is 1.272+0.786j
```

```
# Task 7

# Python Program to convert temperature in celsius to fahrenheit
# change this value for a different result
celsius = 37.5

# calculate fahrenheit
fahrenheit = (celsius * 1.8) + 32
print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit' % (celsius, fahrenheit))
```

```
37.5 degree Celsius is equal to 99.5 degree Fahrenheit
```

```
# Task 8

# Python Program to find the area of triangle

a = 5
b = 6
c = 7

# Uncomment below to take inputs from the user
# a = float(input('Enter first side: '))
# b = float(input('Enter second side: '))
# c = float(input('Enter third side: '))

# calculate the semi-perimeter
s = (a + b + c) / 2

# calculate the area
area = (s*(s-a)*(s-b)*(s-c)) ** 0.5
print('The area of the triangle is %0.2f' %area)
```

```
The area of the triangle is 14.70
```

```
# Task 9

# Solve the quadratic equation ax**2 + bx + c = 0

# import complex math module
import cmath

a = 1
b = 5
c = 6
```

```
# calculate the discriminant
d = (b**2) - (4*a*c)

# find two solutions
sol1 = (-b-cmath.sqrt(d))/(2*a)
sol2 = (-b+cmath.sqrt(d))/(2*a)

print('The solution are {0} and {1}'.format(sol1,sol2))
```

```
The solution are (-3+0j) and (-2+0j)
```

```
# Task 10

# Taking kilometers input from the user

kilometers = float(input("Enter value in kilometers: "))

# conversion factor

conv_fac = 0.621371

# calculate miles

miles = kilometers * conv_fac

print(kilometers, "kilometers is equal to", miles, "miles")
```

```
Enter value in kilometers: 78
78.0 kilometers is equal to 48.466938 miles
```

```
# Task 11

i = 10

if (i > 15):
  print("10 is less than 15")

print("I am Not in if")
```

```
I am Not in if
```

```
# Task 12

i = 20;

if (i < 15):
    print("i is smaller than 15")
    print("i'm in if Block")
else:
    print("i is greater than 15")
    print("i'm in else Block")

print("i'm not in if and not in else Block")
```

```
i is greater than 15
i'm in else Block
i'm not in if and not in else Block
```

```
# Task 13

i = 10
if (i == 10):
    # First if statement
    if (i < 15):
        print ("i is smaller than 15")
    # Nested - if statement
    # Will only be executed if statement above
    # it is true
    if (i < 12):
        print ("i is smaller than 12 too")
    else:
        print ("i is greater than 15")
```

```
i is smaller than 15
i is smaller than 12 too
```

```
# Task 14

i = 20
if (i == 10):
    print ("i is 10")
elif (i == 15):
    print ("i is 15")
elif (i == 20):
    print ("i is 20")
else:
    print ("i is not present")
```

```
i is 20
```